

Principles for Designed-In Security and Privacy for Smart Cities

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ABSTRACT

This paper presents the design and implementation of a process for an exploratory study that identifies a set of principles for designed-in security and privacy for smart city projects from among Global City Teams Challenge (GCTC) – Smart and Secure Cities and Communities Challenge (SC3) participants. The study was conducted based on information from the National Institute of Standards and Technology (NIST) GCTC Action Clusters database and interactions with the project teams. A research process was developed and implemented, comprising the following three steps:

- (1) Investigate project descriptions created by the project leads on the NIST GCTC database and other public sources;
- (2) Gather additional input from volunteer GCTC collaborators; and
- (3) Identify a set of governing principles commonly shared by examples of GCTC projects.

Based on the outcomes of this process, a set of common principles has been identified that enable designed-in security and privacy considerations among the projects: specific technology usage, implementation of a cybersecurity management process and framework, and cybersecurity expertise and public-private partnerships. Characteristics of planning and implementation of security and privacy considerations from four example GCTC projects are described and analyzed in detail to illustrate the process.

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CCS CONCEPTS

• Smart City • Cybersecurity Framework • Cyber-Physical Systems
• Internet of Things • Network Infrastructure

KEYWORDS

Smart city, Cybersecurity, Cyber-Physical Systems, IoT, Infrastructure, Privacy, Resiliency

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1 Introduction

Cities and communities across the US and internationally are attempting to implement community-scale smart systems to improve city management efficiency, increase infrastructure resilience, and provide more convenient and secure services for their constituents [1][2]. These smart city and community initiatives can be complicated projects composed of many different technical domains, including information technology, communication networks, computer analytics, sensors, power electronics, and control systems. These systems and technologies need to interact and communicate in a secure and privacy-protecting manner to effectively operate and meet the needs of city and community stakeholders in terms of privacy and information integrity [3].

Due to numerous interoperability complexities and the lack of in-house cybersecurity expertise, many city and communities are initiating smart city projects in which security and privacy are addressed after initial project development and implementation (rather than considered as an integral component designed-in at the beginning of the project). Although this approach may seem simpler at the beginning, it does not guarantee systematic and holistic implementation of security and privacy that works best for the given smart city project. Moreover, additional patchworks for

security and privacy as an “afterthought” may lead to unintended errors and suboptimal performance of the implemented system. Therefore, it is important to identify a set of governing principles for designed-in security and privacy for smart city projects and take them into account at the beginning of project planning. This paper describes our work in identifying a common set of principles for security and privacy consideration that offer best practices and guidelines that could be used by communities as examples and a template for cybersecurity planning and integration for smart city projects. To illustrate the process, the paper describes several examples of existing smart city projects that are designed and planned from a perspective of trustworthiness, privacy, and security and provide examples that other cities and communities can consider in developing their own plans using available tools such as National Institute of Standards and Technology’s (NIST’s) Cyber-Physical Systems (CPS) Framework [4].

2 Methodology

The methodology to identify a common set of designed-in security and privacy principles followed an investigation of Global City Teams Challenge (GCTC) project descriptions in the NIST GCTC database [5] and discussions with GCTC collaborators. The process used to identify the principles from the technical use cases can be summarized as follows:

- (1) Investigate project descriptions created by the project leads on the NIST GCTC database and other public sources;
- (2) Gather additional input from volunteer GCTC collaborators; and
- (3) Identify a set of governing principles commonly shared by examples of GCTC project.

2.1 Investigating GCTC Database Project Descriptions and other Public Sources

The GCTC is a program initiated and led by NIST, and implemented in partnership with other U.S. federal agencies including the Department of Homeland Security Science and Technology Directorate (DHS S&T). The program was established in 2014 as an outgrowth of the successful SmartAmerica Challenge launched in 2013 [6] to encourage development of replicable and scalable models for incubation and deployment of interoperable, secure, standard-based solutions using advanced technologies, such as Internet of Things (IoT) and CPS, and to demonstrate their measurable benefits in cities and communities. A cornerstone of the program is to incentivize public-private collaborations to develop sustainable integration models that can be used globally by interested community stakeholders [7]. Cities and communities collaborate with technology innovators to develop smart city projects, called Action Clusters, in relation to a chosen sector, with an aim to manage and control city resources in a secure and efficient manner (Figure 1). GCTC Action Clusters cover a wide range of application sectors that cover many infrastructures. Table

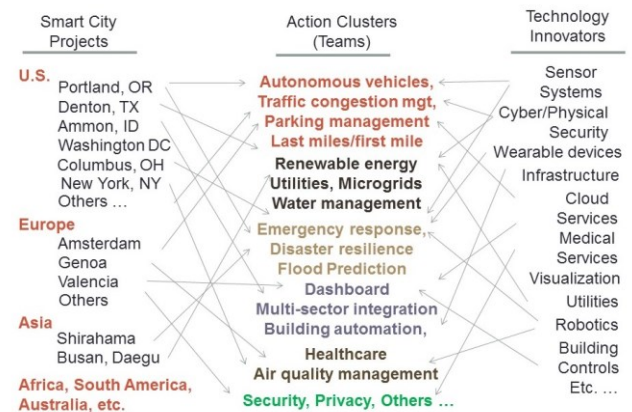


Figure 1. GCTC Action Clusters

1 lists several application sectors covered by GCTC Action Clusters.

GCTC participants are ideal for this type of exploratory case study because they are already engaged in smart city projects and detailed in the GCTC database. Therefore, examples of smart city projects were identified from among the past participants of GCTC and additional information was obtained from other public sources such as the GCTC Wiki [8].

Table 1. GCTC Action Cluster Sectors

Application Areas
Transportation
Public Safety
Utility (Energy, Water, and Environment)
Data Governance and Exchange
Wireless
Education
Agriculture and Rural

2.2 Gathering Additional Input from GCTC Collaborators

The GCTC project descriptions created by the project leads in the database provided an overview of participants’ projects, but in most cases, they offered limited information about cybersecurity integration. Therefore, conversations were needed with participants to fill in the gaps for cybersecurity details that were not discussed in descriptions posted in the NIST database.

To fill in the information gaps about cybersecurity integration, remote meetings were set up with several GCTC teams to explain

the goal of the analysis and gather additional information on their projects.

2.3 Analysis of the Projects and Identifying a Set of Principles for Designed-in Security and Privacy Implementation

The GCTC collaborative partners described their projects in terms of cybersecurity integration as an initial design priority. Summaries of the discussions and project reviews were developed detailing specific information pertinent to relevant concepts and topics for designed-in security and privacy for smart city projects. From the detailed descriptions collected through the investigation of project descriptions on the NIST GCTC database, and one-on-one conversations with the GCTC collaborators, a set of common considerations for designed-in security and privacy could be identified.

3 Smart City Project Use Cases

Four projects are described in this paper as examples of uses cases. The four projects consist of:

- (1) Secure Smart Lights / Sensors project, San Leandro, California, USA;
- (2) Smart Trip Las Vegas: Safer and Connected Transportation, Las Vegas, Nevada, USA;
- (3) Building Portfolio Cyber-Secure, Real-Time Utility Data Integration, Pittsburgh, Pennsylvania, USA; and
- (4) Underground Infrastructure Sensing and Mapping for Smart Maintenance, Sustainability, Usage and Resilience, Burlington Vermont, USA.

3.1 Secure Smart Lights/Sensors, San Leandro, California

The San Leandro project is a utility/energy sector project that is comprised of a secure controllable network of efficient street lights providing reduced energy consumption and safety [9]. The lighting solution offers autonomous or scheduled illumination dimming that can be controlled from the district level down to a specific light node.

The core team, a public-private partnership, consists of a government partner, the City of San Leandro (CSL), and two technology private companies. CSL decided to outsource the project for a developed turn-key solution. The companies collaborated to provide the cybersecurity expertise and the LED based smart lighting solution. The cybersecurity solution adopted by the city is based on the concept of “moving target data protection” designed to render common attacks intractable as it removes contextual cues while increasing obfuscation via various distributed computation techniques [9].

The cybersecurity concerns are handled using the Information Security Maturity Model (ISM3) framework with the Open Source Security Information Management (OSSIM) tool [10][11]. The OSSIM platform implemented provides intrusion detection and prevention.

3.2 Smart Trip Las Vegas: Safer Connected Transportation, Las Vegas, Nevada

The Las Vegas project is a transportation sector project that uses a technology that provides fleet vehicles the capability to gather intelligence about traffic and pedestrian issues to increase pedestrian safety. The project aims to use connected car technologies to provide actionable messaging to drivers on road conditions while adopting open standard vehicle communication protocols based on a consortia-developed technology enabling faster deployment among vehicle manufacturers. The system establishes secure, reliable, two-way wireless connectivity between the in-vehicle unit and city-hosted servers so that vehicle data flows and actionable alerts return to inform the driver. The project also develops analytics and trend reports that support city/region/state transportation planning decisions for the welfare of citizens and visitors [12].

The team, a public-private partnership, consists of two government entities—the City of Las Vegas (CLV) and the Regional Transportation Commission (RTC) of Southern Nevada—and a technology private company.

The two municipal entities provided management and the fleet vehicles. The technology company developed the turnkey solution in terms of software, hardware, integration, and security. The technology was integrated without a cyber-based management plan, but with a security framework leveraged by the private partner and security protocols that were used in the technology.

3.3 Building Portfolio Cyber-Secure, Real Time Utility Data Integration, Pittsburgh, Pennsylvania

The Pittsburgh project is leveraged from an energy efficiency project in the U.S. Department of Energy (DOE) sponsored energy efficiency hub. The foundation of the project focused on plug load technology coupled with utility portfolio data management for energy management and reduction for commercial buildings. Based on visualizations and artificial intelligence algorithms developed at the Carnegie Mellon University (CMU) School of Architecture (SOA), the solution is a cloud-based building information system that can integrate with utility and building data to provide visualizations, benchmarking and recommendations for building portfolio energy efficiencies and indoor environmental quality. This project also plans to include the first independent Smart Building/Smart City training, testing and certification lab at the Energy Innovation Center in Pittsburgh, PA [13].

The core team is a public-private partnership composed of the City of Pittsburgh, a private technology company, and a technology innovation entity.

This project was implemented with no cybersecurity framework, but the private technology company chose and implemented a turnkey plug/Wi-Fi solution with embedded security features.

3.4 Underground Infrastructure Sensing and Mapping for Smart Maintenance, Sustainability, Usage and Resilience, Burlington, Vermont

The Vermont project is a utility-based project for underground infrastructure maintenance and resiliency. This project centers around an autonomous sensor node that can be networked to provide infrastructure monitoring and 3D mapping. The project uses sensing and information technology to determine the state of infrastructure and provide it in an appropriate, timely, and secure format for the users. Information processing techniques convert the data in information-laden databases for use in analytics, graphical presentations, metering and planning [14].

The primary team is a public-private relationship consisting of two cities in Vermont (Burlington and Winooski), a private technology company, and a university.

The Vermont city entities administered overall management and enabled access to city infrastructure, while the private company developed the technical solution. The university researchers addressed security concerns and integrated cybersecurity measures into design decisions, considering the NIST cybersecurity framework.

4 Designed-in Security and Privacy Principles

From the research of the GCTC smart city projects and more extensive examination of a few existing use cases, a set of commonly addressed principles for smart city projects with designed-in security and privacy has emerged: specific technology usage, implementation of a cybersecurity management process, and cybersecurity expertise and public-private partnerships.

4.1 Specific Technology Usage

Most of the projects investigated in this paper clearly focus on a specific technology such as San Leandro's use of moving target data protection technology (Table 2) that is most suitable for a technical use case study rather than a general Internet of Things (IoT) platform. This ensures the security and privacy considerations during planning and implementation of the project can be focused and efficiently managed for specific use cases. Successful examples of specific technology usage can be more readily transferrable and usable by other use cases with similar goals.

Table 2 Specific technologies from example projects

Specific Technologies
Intelligent street light with moving target data protection technology
Wifi smart plug with embedded security features
Infrastructure sensor node considering NIST cybersecurity framework
Traffic monitor with built-in security protocols

4.2 Implementation of a Cybersecurity Management Process and Framework

Many of the projects investigated considered the use of a systematic approach to developing the cybersecurity elements of their smart city/community project, such as San Leandro's efficient street light project. This included following a defined management process (Table 3) and applying a defined framework, such as the NIST Cybersecurity Framework (CSF) used in the Vermont municipalities example, where the framework provides a foundation for understanding and managing security risks (Table 4) [15].

Table 3 Cybersecurity management components

Management
Pre-evaluation
Strategic planning
Operative planning
Implementation
Post-evaluation

Table 4 Core topics of NIST Cybersecurity Framework [15]

NIST Cybersecurity Framework Core Functions
Identify
Protect
Detect
Respond
Recovery

4.3 Cybersecurity Expertise and Public-Private Partnerships

Most of the example use cases relied on cybersecurity expertise from private sector partners to help guide their project. This includes communities who do not have extensive, in-house cybersecurity expertise, such as the City of Pittsburgh and Las Vegas projects, which collaborated with vendors. The inclusion of expert partners from academia or industry to drive cybersecurity design and integration for the project was a major consideration shared by the example use cases. Understanding how such a partnership can work in practice, impacts the productivity of the planning and implementation of security and privacy considerations in these projects.

5 Conclusion

After collaborative discussions with GCTC partners to understand security and privacy aspects of their technology integration and reviewing several projects in the GCTC database, the following common principles and shared best practices were identified for security and privacy consideration and implementation:

- (1) Specific Technology Usage;
- (2) Implementation of a cybersecurity management process and framework; and
- (3) Cybersecurity Expertise and Public-Private Partnerships.

Four example projects were described in this paper to illustrate how these principles and practices can be implemented to develop smart city projects that improve city infrastructure with designed-in security and privacy.

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